ADITYA RASTOGI
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EDUCATION

M.Tech & B.Tech (Hons.), Computer Science and Engineering, IIT Kharagpur, India
Masters GPA: 9.86/10; Graduated: May'21; Thesis: Improvements in Self-Supervised Learning
Bachelors GPA: 9.46/10; Graduated: May'20; Thesis: Deep Learning Visualization

WORK EXPERIENCE

CausalFoundry (team moved from Benshi), Barcelona, Spain Mar'24-Present
Machine Learning Engineer (Go, PostgreSQL, Python)

- Developed a chat interface integrating fine-tuned GPT models with custom function calling for database and RAG queries, deployed in a mobile app, using unstructured for document parsing.
- Implemented contextual bandits for personalized digital health nudges, increasing engagement and in-app spending in production.
- Deployed production models including item2vec recommenders, Recurrent Deep Survival Machines, and hierarchical forecasters (MASE 0.5–0.7) across millions of events.

Benshi.ai, Barcelona, Spain Sept'22-Feb'24
Machine Learning Engineer (Go, PostgreSQL, Python)

- Built a batch data pipeline in PostgreSQL processing 30M+ monthly events and 2500 features, with custom DAG for easy feature expansion without framework lock-in.
- Created synthetic simulation environments for benchmarking and improving reinforcement learning algorithms, contributing results to a KDD workshop paper.

Accenture Japan Ltd., Tokyo, Japan Nov'21-Sept'22
Business and Integration Arch Analyst (Java)

- Automatic test generation in a large codebase by identifying code patterns.
- Meeting with clients and developers, understanding requirements, sprint planning etc.

INTERNSHIPS

Goldman Sachs, Bengaluru, India May'20-Jun'20
Developed an end-to-end system to classify tickets received through emails, using multiple classifiers.

University of British Columbia, Vancouver, Canada May'19-Jul'19
Pattern Matching in Trillion Edge Graphs; *Advisor: Prof. Matei Ripeanu*
Maximized lateral work reuse for pattern matching in graphs in a distributed systems setting.

University of Sydney, Camden, Sydney, Australia Dec'18-Jan'19
Facial Landmarks Detection; *Advisor: Dr. Mehar Khatkar*
Implemented a deep learning model for detecting facial landmarks to monitor aquariums.

IIT Kharagpur – Sponsored by Shell India Pvt. Ltd. May'18-July'18
Topic: Sensor Diagnostics; *Advisor: Prof. Swanand Khare*
Used gaussian-mixture models to detect change points in data from sensors in a chemical plant.

Projects

- **Discourseify** (A social media platform with conversational recommenders to facilitate debates) (*~5k loc in Go, PostgreSQL and Vue.js*)
- **Lexica** (A vocabulary management tool using LLMs) (*~2.5k loc in Go, SQLite and Vue.js*)
- **AlphaZero from scratch in PyTorch** (*~1.2k loc in Python*)
- **Switch Transformers from scratch in PyTorch** (*~800 loc in Python*)
- **Self-Supervised Learning in Computer Vision** (SimCLR, Moco-V2) (*~700 loc in Python*)
- **Saliency Map Extraction in PyTorch** (*~100 loc in Python*)
- **Automatic differentiation engine** (*~1000 loc in Go*)
- **Evolutionary Algorithm to train a self-driving car on a racetrack** (~1000 loc in JavaScript)
- **File system implementation using File Allocation Table and Index Nodes** (~2000 loc in C++)
- **Implementation of a Simplified File Transfer Protocol** (~1000 loc in C)
- **Implementation of Cryptographic Algorithms** (~2000 loc in C)
- **Map-Reduce from scratch using gRPC in Python** (~450 loc in Python)
- **Off-Policy Monte Carlo Control in Reinforcement Learning** (~500 loc in Python)
- **Policy Iteration in Reinforcement Learning** (~200 loc in Python)

Publications

Olaniyi, B. et al. (2025) **Data-Driven Approach to Capitation Reform in Rwanda**, arxiv preprint
Xu, Jiamin et al. (2025) **From Restless to Contextual: A Thresholding Bandit Approach to Improve Finite-horizon Performance,** arxiv preprint
Pandey, A. et al. (2024) **Darsi: A deep auto-regressive time series Inference architecture for forecasting of aerodynamic parameters**, Journal of Computational Science, 82, p. 102401. doi:10.1016/j.jocs.2024.102401.
del Río, A.F. et al. (2024) **Adaptive user journeys in pharma E-commerce with reinforcement learning: Insights from SwipeRx**, KDD CJ workshop
del Río, A.F. et al. (2024) **Adaptive behavioral AI: Reinforcement learning to enhance pharmacy services**, KDD AIBS workshop
Periáñez, Á. et al. (2024) **Optimizing HIV patient engagement with reinforcement learning in resource-limited settings**, KDD epidamik workshop
Periáñez, Á. et al. (2023) **The Digital Transformation in Health: How AI can Improve the Performance of Health Systems**, Harvard TH Chan School of Public Health, Takemi Program in International Health
Rastogi, A. et al. (2023) **Synthetic Data Generator for adaptive interventions in Global Health**, ICLR MLGH workshop

Skills And Expertise

Languages: **Go**, **SQL** (PostgreSQL, SQLite, ClickHouse), **Python**, JavaScript (p5.js, Vue.js), C, C++, Java, R, Stata, MATLAB, HTML5, CSS
Libraries: **PyTorch**, **Numpy**, **Scipy**, **Matplotlib**, **Darts**, JAX, gRPC, Keras, TensorFlow

Others

- **Teaching Assistant – Machine Learning (Spring 2021), Algorithms-II (Autumn 2020)**
- Fluent in English (TOEFL: 112/120) and Hindi, beginner in Japanese (~ N4) and Spanish (~A1).